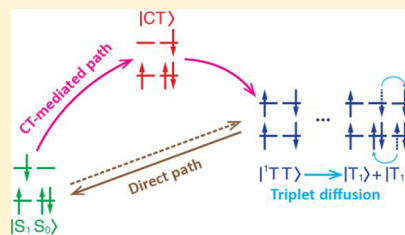


Distinct Routes of Singlet Fission and Triplet Fusion: A Fluorescence Kinetic Study of Rubrene

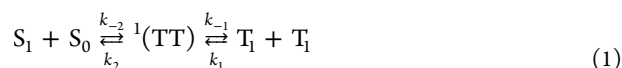
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Supporting Information

ABSTRACT: Singlet fission of organic molecules has attracted recent attention owing to its potential advantages in organic photovoltaic and electroluminescence applications. Its microscopic mechanism however remains stymied. Large couplings from charge-transfer (CT) state mediation were invoked to explain the ultrafast singlet fission rate observed in crystalline polyacene, but its experimental confirmation is still lacking. The singlet fission and triplet fusion of amorphous rubrene were investigated with time-resolved photoluminescence spectroscopy at different temperatures to extract the rates of singlet fission, triplet fusion, and triplet hopping. On the basis of the Marcus electron-transfer model, the deduced electronic coupling constant of the singlet fission process was found to be larger than that of the triplet fusion process, indicating that the singlet fission process undertakes a CT-state-mediated channel while the triplet fusion process assumes a direct channel. This study thus confers supporting evidence of the existence of the CT-state-mediated channel for singlet fission of rubrene and offers an experimental approach to study singlet fission dynamics.



Splitting one singlet exciton into two triplet excitons—a process known as singlet fission—has been proposed as a possible pathway to increase photovoltaic efficiency above the Shockley–Queisser limit.¹ There have been scores of experimental and theoretical efforts to determine the rates of singlet fission of different materials.^{2–5} An agreement between the two efforts is yet to be reached. A kinetic scheme describing the singlet fission and triplet fusion processes is traditionally written as¹



where the singlet exciton (S_1) splits into two separate triplet excitons ($T_1 + T_1$) via the intermediate correlated triplet pair ${}^1(TT)$ that also mediates the reverse triplet fusion process. The singlet fission rate, k_{-2} , depends on the electronic coupling involved and the available thermal energy if the process is endothermic.⁴ Two coupling models have been proposed: (1) direct coupling between $|S_1 S_0\rangle$ and $|{}^1TT\rangle$ and (2) indirect coupling mediated by a charge transfer (CT) state. Theoretical calculations favor the mediated route because the much larger coupling strength is in line with the ultrafast fission rates measured for pentacene,⁶ but direct experimental evidence is still lacking.

Subsequent to the fission event, the triplet pair can either fuse back into a singlet exciton or separate into two uncorrelated triplet excitons. The direct reverse process of singlet fission is called geminate triplet fusion, while its nongeminate counterpart is played by two encountered triplet excitons out of diffusion and normally takes place in intense photoexcitation or in confined geometries (e.g., thin solid-state film). Both singlet fission and geminate triplet fusion are affected by identical electronic coupling in the direct pathway, while the indirect pathway mediated by the CT state suggests that these two processes might proceed along different energetic pathways. In this study, we extracted the singlet fission and geminate triplet fusion rates (k_{-2} and k_2 , respectively) of rubrene (5,6,11,12-tetraphenylanthracene) with time-resolved photoluminescence (TRPL) spectroscopy, followed by a comparison of the respective coupling factors extracted from their temperature dependences, to differentiate the two singlet fission pathways. Two measures were taken to simplify the retrieval of k_{-2} and k_2 . First, the contribution from nongeminate triplet fusion (the k_1 process) was suppressed as the photoexcitation intensity was kept small enough so that the

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excited singlet excitons were well separated and the non-generative fusion of isolated triplet excitons—produced via singlet fission and subsequent hopping—took place on a much longer time scale. Second, the dynamics of the exciton occurred in an unconfined environment (such as a thick sample) so that the built-up exciton distribution owing to constrained diffusion was negligible. Accordingly, the rate k_{-1} can be treated as the triplet hopping rate; k_2 is the geminate triplet fusion rate; k_1 effectively approaches zero (section S1, Supporting Information).⁷

Energy states pertinent to excitonic species of a thermally deposited amorphous rubrene film were characterized by steady-state absorption and photoluminescence (PL) spectroscopy. The PL spectra (Figure 1a) show that the highest-energy

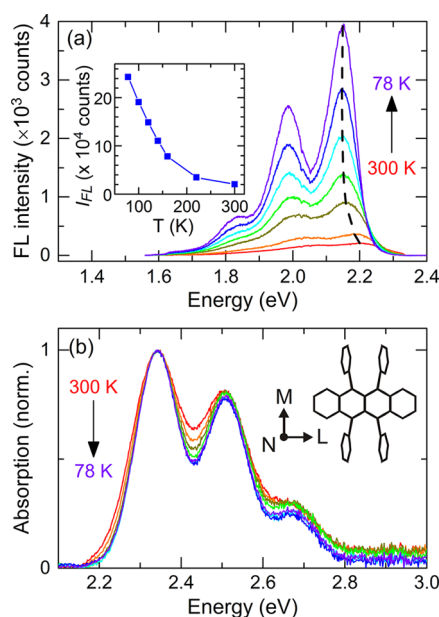


Figure 1. (a) Temperature-dependent fluorescence (FL) spectra. The dashed line is a guide to the eye to illustrate the peak shift. Inset: total FL intensity (I_{FL}) plotted against temperature (T). (b) Normalized temperature-dependent absorption spectra. Inset: rubrene molecular structure.

vibronic peak is red-shifted from 2.2 to 2.15 eV as the temperature is lowered from 300 to 140 K and remains at 2.15 eV for lower temperatures (from 140 to 78 K), while the PL intensity increases with the decrease in temperature monotonically. On the other hand, there is no spectral shift in the corresponding absorption spectra (Figure 1b), indicating that the temperature-dependent dielectric effect and additional intermolecular interaction owing to thermal contraction are negligible. The temperature-dependent PL spectra in Figure 1a indicate that there are at least two emitting species: the species dominating the red-shifted narrow spectrum at 78 K (Species *a*) is more quenched than the one mainly contributing to the broad spectrum at 300 K (Species *b*). On the basis of an estimation from the integrated PL intensity that characterized Species *a* at 78 K and Species *b* at 300 K (Figure 1a, inset), less than 10% of the total population is Species *b*. Further information can be provided by time-gated PL spectra. The PL spectra of the rubrene film at 78 and 300 K obtained with their respective time-gated durations (0–5 ns, 10–110 ns, 110–310 ns, 0.3–0.8 μs , and 0.8–4.8 μs for 78 K; 0–5 ns, 40–540 ns, and 1–4.5 μs for 300 K) are shown in Figure 2. The PL

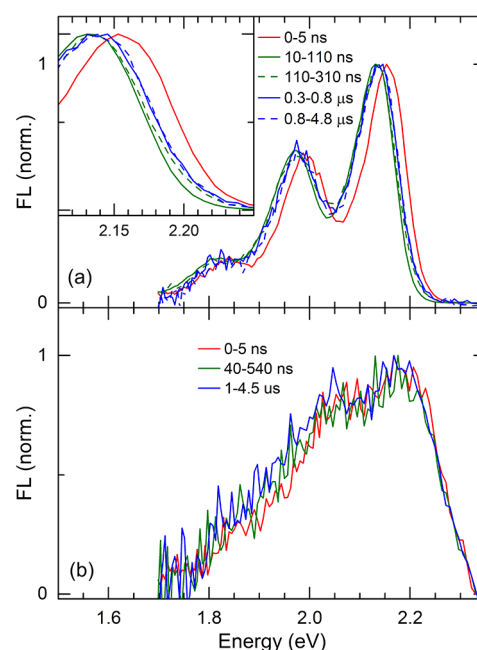


Figure 2. Time-gated PL spectra recorded at (a) 78 and (b) 300 K. The inset highlights the peak shifts near the PL peak maximum.

spectrum at 78 K taken during the initial 5 ns approximates the steady-state PL spectrum at 78 K shown in Figure 1a and gives the highest-energy vibronic peak at 2.15 eV; that of the PL spectrum acquired between 10 ns and 0.3 μs resides at 2.13 eV; finally, that of the PL spectrum acquired between 0.3 and 4.8 μs is located at 2.14 eV. Namely, the time-gated PL spectra between 0 and 4.8 μs and the steady-state PL spectrum at 78 K approximately share the same vibronic feature, thus signifying an emitting species capable of fission—Species *a*. On the other hand, all three time-gated PL spectra at 300 K are almost identical to the corresponding steady-state PL spectrum shown in Figure 1a—conferring no vibronic feature, reflecting an emitting species without PL quench by fission—Species *b*, which is likely the molecules with unfavorable neighbors (e.g., a larger intermolecular separation and/or undesirable orientation) for the formation of $|^1\text{TT}\rangle$. One evidence of the assignment of Species *b* is that its PL spectrum at 300 K is similar to that of isolated rubrene molecules in toluene (Figure S3 in the SI) owing to inhomogeneous twisting motions of the rubrene backbone and attached phenyl groups in solvent.⁸ Furthermore, the absent vibronic feature of the prompt and delayed PL spectra at 300 K indicates that the PL of Species *a* is efficiently quenched at room temperature and the S_1 state of Species *b* is nonuniformly distributed, again consistent with its weak interaction with surrounding molecules. The well-resolved vibronic feature of the PL spectra of Species *a* further affirms its uniform and closer molecular packing, thus resulting in efficient singlet fission between adjacent molecules. Besides, the closer packing of the Species *a* exciton supports the red shifting in the PL spectra with the decrease in the temperature, shown in Figure 1a. The assignment of the singlet species paves the foundation for the interpretation of the time-gate PL spectra.

A dynamical scenario emerges based on the experimental results of Figures 1 and 2. Upon photoexcitation, the singlet excitons generated can be separated into two kinds: Species *a* and *b* excitons. The former one dominates the ensemble of the

generated excitons and can undergo efficient singlet fission, while the latter one holds a smaller amount and decays less via the fission process. At 300 K, the thermal energy prompts the singlet fission process, leaving mostly Species *b* excitons. On the other hand, at 78 K, the singlet fission process is not efficiently activated, thus maintaining the population of the Species *a* exciton even at long time delays. Although the generated triplet exciton pairs can undertake the geminate fusion process, the singlet excitons are reformed at either the original molecule or the adjacent one and thus manifests as its original species, portraying their respective PL signatures at long time delays. Namely, the population dynamics of these two species evolve independently. Should the exciton density be high enough to enable the nongeminate fusion process, the population dynamics of the two species would be coupled, thus complicating the evolution of the PL spectra. Besides the main dynamic characteristics of the scenario, there exists some smaller details. Note that the PL spectrum between 0 and 5 ns at 78 K is red-shifted with respect to those after 10 ns, shown in Figure 2a, indicating that the Species *a* exciton can be further separated into two subtypes: Species *a*₁ and *a*₂. The former subtype manifests a bluer PL feature and faster singlet fission process than the latter one because the bluer PL feature present at the time gate between 0 and 5 ns is replaced by the relatively red PL feature at the time gates after 10 ns.

The dynamics of the singlet fission and geminate triplet fusion processes are portrayed in the time-resolved PL behaviors at different temperatures (Figure 3). Note that the

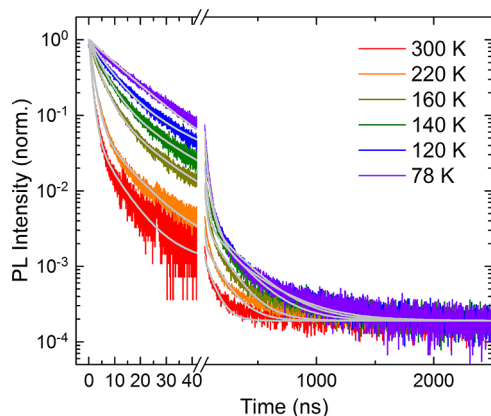


Figure 3. PL decay of an amorphous rubrene thin film as a function of temperature. A break in the *x*-axis emphasizes the short and long time regimes. The excitation densities are kept to $9 \times 10^{15} \text{ cm}^{-3}$ at 300 K and $2 \times 10^{17} \text{ cm}^{-3}$ at other temperatures to suppress contribution of nongeminate triplet fusion in delayed PL. Gray lines are the global fitting results from a physical model described in the text.

higher the temperature is, the faster the integrated PL signal decays, and it appears that there exist multiple decay rates in these transient PL curves. To interpret the data and, more specifically, to extract the pertinent factors that govern the processes, a kinetic model that quantitatively reflects the corresponding mechanisms involved needs to be developed. On the basis of the kinetic scheme, illustrated in eq 1, a set of coupled rate equations, considering only geminate fusion, is formulated

$$dN_{S1}/dt = -(k_r + k_{-2})N_{S1} + k_2N_{TT} \quad (2)$$

and

$$dN_{TT}/dt = k_{-2}N_{S1} - (k_2 + k_{-1})N_{TT} \quad (3)$$

where N_{S1} and N_{TT} are the populations of singlets and geminate triplet pairs, respectively, and k_r is the singlet radiative decay rate. The analytic solution for N_{S1} is given as

$$N_{S1} = Ae^{-\alpha_1 t} + Be^{-\alpha_2 t} \quad (4)$$

where

$$\alpha_1 = (R + Q)/2 \quad (5a)$$

$$\alpha_2 = (R - Q)/2 \quad (5b)$$

$$A = N_{S1}^0[1 + (R/Q)]/2 \quad (5c)$$

$$B = N_{S1}^0[1 - (R/Q)]/2 \quad (5d)$$

$$R = k_r + k_{-2} + k_2 + k_{-1} \quad (5e)$$

$$Q = \{R^2 - 4[k_r k_2 + k_{-1}(k_r + k_{-2})]\}^{1/2} \quad (5f)$$

and N_{S1}^0 is the initial population at S_1 . According to eq 4, the PL signal—being proportional to the population at S_1 (N_{S1})—decays biexponentially with one prompt (α_1) and another delayed (α_2) rate, where $\alpha_1 > \alpha_2$. They are both functions of the three involved kinetic rates. The retrieval of the three kinetic rates can therefore be undertaken readily with use of eqs 4 and 5 from the PL transient behaviors of amorphous rubrene. For a homogeneous organic system that portrays single singlet fission–triplet fusion dynamics, eqs 4 and 5 indicate that there should be only two decay rates in the observed PL signal. On the other hand, for a heterogeneous system such as amorphous rubrene, the PL signal will display decay with more than two rates because different species may confer dissimilar decay dynamics.

As discussed above, although there exist two fission-capable singlet exciton species (Species *a*₁ and *a*₂) and one fission-incapable species (Species *b*) of amorphous rubrene in low excitation condition from temperature-dependent steady-state and time-gated PL spectra (Figures 1 and 2, respectively), Species *b* was not considered in the analysis owing to its negligible contribution. Accordingly, a linear combination of the individual dynamics of Species *a*₁ and *a*₂, governed by eqs 4 and 5, is fitted globally to the temperature-dependent decay curves in Figure 3. Their respective physical rates extracted are plotted in Figure 4. The singlet fission rates extracted at 300 K, $\sim 1 \text{ ns}^{-1}$, are comparable to those from other reports on amorphous rubrene.^{5,9} Further details on the fitting procedures and the numerical values of the extracted rates can be found in the Supporting Information. Note that the singlet fission rate of Species *a*₁ is comparable with that of Species *a*₂ over the whole temperature range, i.e., $k_{-2,a1} \approx k_{-2,a2}$, while the geminate triplet fusion rate of Species *a*₁ is larger than that of Species *a*₂, i.e., $k_{2,a1} > k_{2,a2}$. Furthermore, $k_{-2,i}$ has a more sensitive temperature dependence than $k_{2,i}$ does. On the other hand, the triplet hopping rate of Species *a*₁ is larger than that of Species *a*₂, i.e., $k_{-1,a1} > k_{-1,a2}$. Both singlet fission/triplet fusion ($k_{-2,i}$ and $k_{2,i}$) and triplet exciton transfer (or called triplet hopping) ($k_{-1,i}$ and $k_{1,i}$) can be modeled by the Marcus electron-transfer equation provided that the coupling involved is weak so that the pertinent states can be treated as nonadiabatic states^{4,6,10}

$$k = AT^{-1/2}e^{-\Delta G^*/k_B T} \quad (6)$$

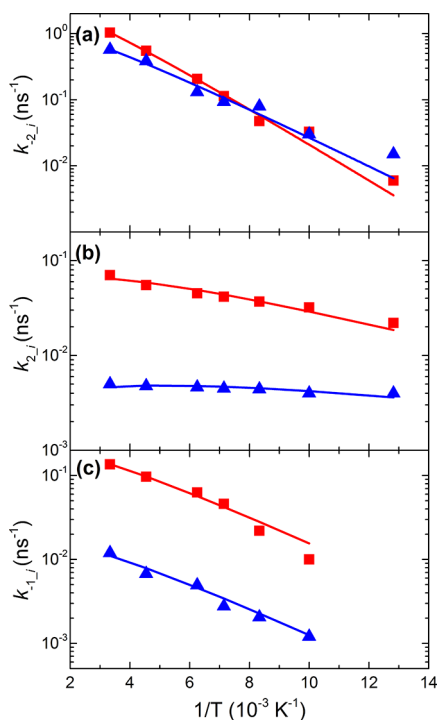


Figure 4. Temperature-dependent behaviors of (a) singlet fission, $k_{-2,ij}$, (b) triplet fusion, $k_{2,ij}$, and (c) triplet hopping, $k_{-1,ij}$ rates. The red square and blue triangle symbols represent rates extracted from Species a_1 and a_2 , respectively. The data points were fitted by the Marcus equation, eq 6.

with

$$A = \frac{2\pi}{\hbar} V^2 \frac{1}{\sqrt{4\pi\lambda k_B}} \quad (7a)$$

and

$$\Delta G^* = \frac{\lambda}{4} \left[1 + \frac{\Delta G^0}{\lambda} \right]^2 \quad (7b)$$

where k_B and $\hbar$ are the Boltzmann and reduced Planck constants, respectively, and V is the electronic coupling between the initial and final states. However, the two electron transfer events—singlet fission and splitting of the correlated triplet exciton pair—differ in one fundamental aspect: singlet fission and geminate triplet fusion take place between $|S_1S_0\rangle$ and $|^1TT\rangle$, while splitting and reformation of the correlated triplet exciton pair occur between $|^1TT\rangle$ and $|T_1S_0\rangle$. Namely, the electron-transfer event in the former case takes place via reorganization of the energetics and spin state of electrons within a fission-capable molecular pair. On the other hand, the event in the latter case occurs between one of surrounding unexcited molecules and is well described by Dexter energy transfer. The relevant energies involved in these two electron-transfer events are depicted in Figure 5. Fitting the temperature-dependent behaviors in Figure 4 with eq 6 yields the respective activation energies, Gibbs free energies, electronic coupling constants, and reorganization energies, shown in Table 1. Note that because $\Delta G^0 = 0$ for triplet hopping, $\Delta G^* = \lambda/4$. The extracted fission activation energies from both species, $\Delta G_{2,ij}^*$, agree with those from other reports on crystalline rubrene^{11,12} but are larger than those obtained from amorphous rubrene.^{9,13} The comparable fission and

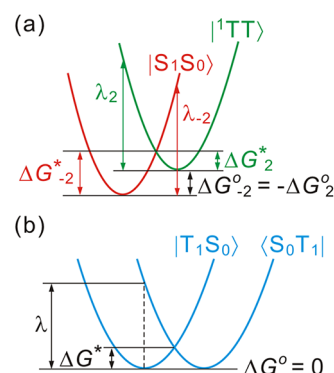


Figure 5. Energy-state schematics of (a) singlet fission and (b) triplet hopping. The activation energy, ΔG^* , reorganization energy, λ , and Gibbs free energy, ΔG^0 , are shown.

Table 1. Derived Energetic Parameters for Molecular Species a_1 and a_2 ^a

	ΔG^* (meV)	λ (meV)	ΔG^0 (meV)	V (μ eV)
$k_{-2,a1}$	58 (2)	138 (6)	40 (5)	154 (11)
$k_{2,a1}$	17 (1)		−40 (5)	53 (2)
$k_{-1,a1}$	35 (1)	141 (5)	0	77 (3)
$k_{-2,a2}$	47 (2)	95 (9)	39 (6)	85 (8)
$k_{2,a2}$	8 (8)		−39 (6)	11 (5)
$k_{-1,a2}$	35 (1)	141 (5)	0	22 (3)

^aActivation energy, ΔG^* ; reorganization energy, λ ; Gibbs free energy, ΔG^0 ; electronic couplings, V . Error bars are shown in parentheses. The error bars for ΔG^* are fitting errors. All other error bars are propagated error from the derivation.

fusion activation energies between the two species and the equivalent values for ΔG_i^0 indicate that their physical configurations do not differ significantly. The reorganization energy derived from triplet hopping, $\lambda_{-1} = 141$ meV, is comparable to 145 meV deduced for isolated rubrene from photoelectron spectroscopy¹⁴ and other theoretical estimates,¹⁵ indicating the applicability of separating the contribution of the reorganization energy into geometrical relaxation in individual rubrene molecules.

As stated in the introduction, $|S_1S_0\rangle$ and $|^1TT\rangle$ can be coupled via a direct or mediated pathway. If the singlet fission occurs via a mediated pathway, the returning triplet fusion can take place via a direct coupling process and the coupling constant of singlet fission can be different from those of triplet fusion. On the other hand, should the singlet fission proceed along a direct channel, it shares the same coupling constant with triplet fusion. On the basis of the extracted results listed in Table 1, the singlet fission coupling constants, $V_{-2,ij}$, of the two species are significantly larger than the triplet fusion coupling constants, $V_{2,ij}$. We emphasize that rate favoring of singlet fission over triplet fusion by the nine possible spin states of $|^1TT\rangle$ has been explicitly incorporated in the derivation;¹ hence, the difference between the coupling constants points directly to the different underlying quantum mechanical pathways. Theoretical calculations with an optimized crystal structure showed that the mediated coupling can be 1–2 orders of magnitude larger than direct coupling.⁶ An important factor that affects the strength of mediated coupling is the energy difference term, ΔE_{CT} , that is, the energy difference between the mediating charge transfer state, $|CT\rangle$, and the states $|S_1S_0\rangle$ and $|^1TT\rangle$.¹⁶ ΔE_{CT} appears in the denominator; thus, as ΔE_{CT}

increases, the strength of mediated coupling originating from the particular CT state diminishes. Because the molecular conformation of amorphous rubrene is more relaxed compared to that of crystalline rubrene, its CT state energy is higher and away from the energy of $|S_1S_0\rangle$ and $|^1TT\rangle$ due to the increased molecular separation. Accordingly the larger ΔE_{CT} explains why a smaller enhancement from mediated coupling is observed in our amorphous rubrene sample. Triplet hopping via two simultaneous electron transfers is a directly coupled process; the comparable coupling constants derived for k_2 and k_{-1} in both species thus further supports that triplet fusion proceeds via a direct coupling. As a final note, the doubling of the triplet hopping rates was accounted for in the derived coupling constant for triplet hopping due to possible pair dissociation via either triplet.

In sum, temperature-dependent steady-state and TRPL experiments were carried out on an amorphous rubrene film to investigate its singlet fission dynamics. The excitation intensity was carefully adjusted to ensure that the nongeminate triplet fusion process was absent in the acquired experimental results, thus simplifying the data analysis. The change in the steady-state PL characteristics as a function of temperature indicates that there are two dominant fission-capable singlet species and one fission-incapable singlet species. The assignment was supported by the time-gated PL spectra. The rate equations for the singlet and triplet pair under negligible nongeminate triplet fusion were derived and solved. The analytic solutions obtained were used to fit the PL decays at different temperatures. The extracted rates of singlet fission, geminate triplet fusion, and triplet hopping were analyzed with the Marcus electron-transfer model. The fact that the extracted electronic coupling constants of singlet fission and triplet fusion are different supports the mediated coupling channel for singlet fission and the direct coupling channel for triplet fusion. The larger coupling constant for singlet fission has a significant contribution from CT-state-mediated coupling, whereas such a coupling channel vanishes in triplet fusion with only direct coupling remaining. The implication of this finding suggests that an optimized molecular design favoring singlet fission for photovoltaic applications may not necessarily be optimized for triplet fusion in organic LED applications.

EXPERIMENTAL METHODS

Sample Fabrication. The rubrene sample was prepared in a vacuum chamber with a base pressure of $<8 \times 10^{-6}$ Torr. A glass substrate was cleaned by acetone and isopropyl alcohol, each for 10 min, before being installed in the chamber. A 80 nm LiF layer was first thermally deposited on the cleaned glass substrate, followed by thermal deposition of a 100 nm rubrene thin film. The LiF layer isolates any possible exciton quenching channel from the glass substrate. After thin film fabrication, the sample was directly transferred to the glovebox (O_2 and $H_2O < 1$ ppm) for the encapsulation process with a cover glass and UV sealant. For the 100 nm rubrene thin film, no obvious peak was found in the X-ray diffractograph, indicating its amorphous characteristic.

Optical Characterization. All optical signals were recorded through a cryostat window with a 10 $\times$ objective (MPLFLN-BD, Olympus) in epifluorescence configuration. The samples were excited by a frequency-doubled 6 ps, 76 MHz mode-locked Nd:VAN laser, emitting at 532 nm; the repetition rate was reduced to 150 kHz with a pulse picker (pulseSelect, APE) to ensure that all of the excited species returned back to their

ground state before the next excitation pulse. Collected signal were dispersed by a 14 cm monochromator (microHR, Horiba Scientific) equipped with a 600 gr/mm grating, subsequently detected by a cooled CCD (Newton, Andor) or a hybrid PMT (HPM-100-40, Becker & Hickl). The decay dynamics were recorded with two time windows (50 ns and 5 μ s) in a time-correlated single-photon counting (TCSPC) setup. The two data sets were combined by multiplying the photon counts collected with the 50 ns time window by a factor of 100 to account for the difference in the time bins. The full width at half-maximum of the instrument response function (IRF), as shown in Figure S1a, was 123 ps, which is much smaller than the fastest prompt fluorescence decay. Because the decay lifetime extracted with or without convolution of the IRF differed by less than 2%, IRF is not included in the fittings for simplicity. The time-resolved fluorescence spectra were recorded through a gated photon counter (SR400, SRS) with the monochromator in scanning mode. The absorption spectra were recorded with a halogen lamp impinging from underneath the cryostat through the same optical path as the fluorescence measurements. The beam passing through the sample without rubrene layer was used as the reference. The absorption spectra were normalized to the absorption peak.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b08677.

Excitation power dependence, species assignment, solution spectrum, energy transfer, fitting procedures, and extracted rate constants (PDF)

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Notes

The authors declare no competing financial interest.

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